

QVAC Local Upscaler

Reviewer guide — how to verify this submission, step by step

Repo: <https://github.com/geographics0714/qvac-local-upscaler>

X post: <https://x.com/Geo344861554566/status/2100462023566139791>

What it does: runs Tether's QVAC SDK `upscale()` (RealESRGAN x4plus-anime-6B) to super-resolve a photo entirely on the reviewer's own machine — no cloud call, no API key.

Part 1 — Inspect the repo (no install needed)

These checks can be done directly on GitHub, in under two minutes, and map one-to-one to the moderator checklist.

#	Check	Where to look	What you'll see
1	Public, open-source license	Repo homepage	"Public" badge + "MIT license" badge under the About section
2	@qvac/sdk declared $\geq$ 0.19.0	package.json, line 23	"@qvac/sdk": "^0.19.0"
3	Calls loadModel + a real AI function	src/upscale.js, lines 10–15 (import) and 139, 157	Imports loadModel/upscale from "@qvac/sdk"; calls loadModel(...) then upscale({...})
4	No cloud AI service	src/upscale.js, full file	Only imports are node:fs, node:path, node:url, node:child_process, and @qvac/sdk — no openai/anthropic/gemini/replicate anywhere
5	Not a fork/copy of qvac-examples	Commit history + file layout	Single custom CLI file, original comparison-page generator, no shared file names with tetherto/qvac-examples
6	$\geq$ 3 commits authored by submitter	Commits page	8 commits, all authored by GeorgeC0714
7	X post links repo + tags @qvac	X post URL above	Post text: "Built a local AI upscaler with @qvac ... " + repo link
8	SDK calls actually exist	Compare against @qvac/sdk v0.19.x API docs	loadModel, upscale, unloadModel, REALESRGAN_X4PLUS_ANIME_6B all appear in the official API Summary

Part 2 — Run it yourself (optional, ~5 minutes)

This confirms the code doesn't just read correctly — it actually performs on-device inference.

Requirements

- Node.js $\geq$ 22.17
- Windows with Vulkan $\geq$ 1.4 (or macOS/Linux with Metal/Vulkan — see the repo's System requirements link)

- ~2 GB free disk space (model weights download on first run only, ~18 MB for this model)

Step 1 — Clone the repo

```
git clone https://github.com/geographics0714/qvac-local-upscaler.git
cd qvac-local-upscaler
```

Step 2 — Install dependencies

```
npm install
```

Watch the output: it pulls @qvac/sdk from the public npm registry, nothing else unusual.

Step 3 — Run the smoke test (bundled sample image, no photo needed)

```
node src/upscale.js samples/test-gradient.png
```

Expected terminal output, in order:

- ■ Loading RealESRGAN model on-device (first run downloads the weights)...
- A download progress bar the first time only (e.g. ■ Downloading 100% (17.9/17.9 MB)) — skipped on later runs once cached locally
- ■ Model loaded. Running on-device upscale...
- A stats: {...} line — check for "backendDevice": "gpu" (or "cpu") and millisecond-scale timings. This is the strongest single proof of on-device execution: there is no network call in the inference path, and cloud APIs cannot return in low hundreds of milliseconds with these numbers.
- ✓ Wrote upscaled image to ...test-gradient.upscaled.png
- ✓ Wrote before/after comparison page to ...comparison.html

Step 4 — View the result

The script auto-opens comparison.html in your default browser. You'll see the original 32×32 gradient PNG next to a sharper 128×128 output, generated by upscale(). If it didn't open automatically, open the file manually from samples/comparison.html.

Step 5 (optional) — Try your own photo

```
node src/upscale.js path\to\any-photo.jpg
```

Any PNG/JPEG works. Larger or more detailed photos make the upscaling difference easier to see.

Reference: how the code calls the SDK (src/upscale.js, lines 139–171)

```
const modelId = await loadModel({
  modelSrc: REALESRGAN_X4PLUS_ANIME_6B,
  modelType: "diffusion",
  modelConfig: { mode: "upscale", upscaler: { tile_size: args.tileSize } },
});

const { outputs, stats } = upscale({ modelId, image: inputBuf, repeats: args.repeats
```

```
});  
const [upscaledPng] = await outputs;  
  
await unloadModel({ modelId });
```

Questions or anything CANNOT VERIFY? The full source is one file: `src/upscale.js` in the repo above.